Calculus I

Computation problems #9

Topics:

- Basic integral formulas
- Integration by parts
- Trigonometric integrals

Problem 1. $\displaystyle\int \frac{x^2}{x^2+1}\,dx$

Problem 2. $\displaystyle\int_{\pi/4}^{\pi/3} \frac{dx}{\cos^2 x \tan x}$

Problem 3. $\displaystyle\int \frac{dx}{x-\sqrt{x}}$

Problem 4. $\displaystyle\int \frac{2^{\ln z^3}}{16z}\,dz$

Problem 5. $\displaystyle\int x\cos x\,dx$

Problem 6. $\displaystyle\int x^2 e^x\,dx$

Problem 7. $\displaystyle\int e^x \cos x\,dx$

Problem 8. $\displaystyle\int \cos^5 x\,dx$

Problem 9. $\displaystyle\int_0^{\pi/4} \sqrt{1+\cos 4x}\,dx$

Problem 10. $\displaystyle\int \tan^4 x\,dx$

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Answer to selected problems

5. $x\sin x + \cos x + C$

6. $x^2e^x - 2xe^x + 2e^x + C$
7. $\dfrac{e^x \sin x + e^x \cos x}{2} + C$
8. $\sin x - \dfrac{2}{3}\sin^3 x + \dfrac{1}{5}\sin^5 x + C$
9. $\dfrac{\sqrt{2}}{2}$
10. $\dfrac{1}{3}\tan^3 x - \tan x + x + C.$